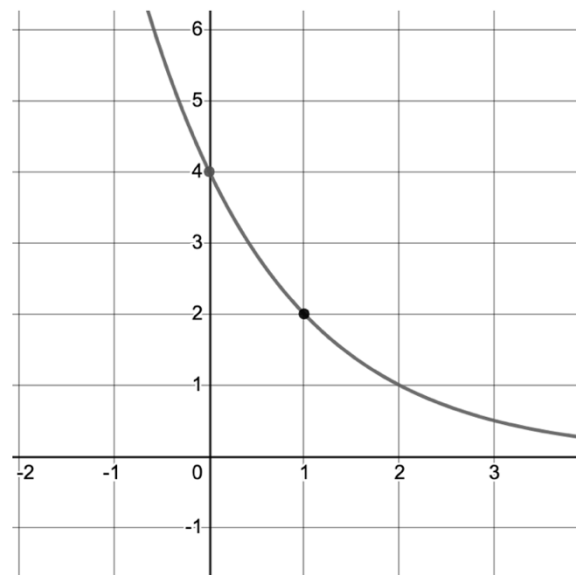


Algebra 1
Unit 11
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

The graph of an exponential function is shown.



1. Find the value of a by substituting the point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{4} = a(b)^{\boxed{0}} \quad a = 4$$

2. Use the value of a found in question 1 and the point $(1, 2)$ to determine the value of b .

$$y = 4(b)^x$$

$$\frac{2}{4} = \frac{4(b)^1}{4} \quad b = \frac{1}{2}$$

3. Write the function for the graph in both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$
$f(x) = 4\left(\frac{1}{2}\right)^x$	$f(x) = 4(1 - 0.5)^x$

The table of an exponential function is shown.

x	0	1	2	3	4
$f(x)$	0.4	1.2	3.6	10.8	32.4

4. What is the initial value, a ?
 $a = 0.4$
5. How does the value of $f(x)$ change as the x -value increases by 1?
As the x -value increases by 1, the y -value or $f(x)$ value is multiplied by a factor of 3.
6. Write the function in the form $f(x) = a(b)^x$
 $f(x) = 0.4(3)^x$

For questions 7-10, write an equation from the table or graph by finding the initial value, a , and common ratio, b .

